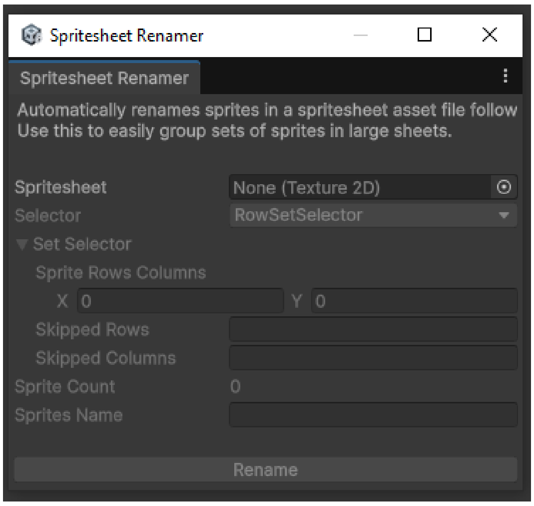
Sprite Tools Documentation

Sprite Tools is a set of 3 Unity Editor Tools meant to speed up workflows for 2D Spritesheet-heavy games. After installation, all tools can be found in the Unity toolbar under Tools > Sprites, and are named “Animator Override Controller Maker”, “Spritesheet Animation Clip Maker”, and “Spritesheet Renamer”. The tools are designed to provide a continuous workflow.

Spritesheet Renamer

Spritesheet Renamer is a tool that allows the user to quickly rename all the sprites in a single Texture2D sheet following a user-defined pattern. When a Texture2D is imported and set up as a Sprite containing multiple Sprites, Unity applies a default naming scheme “TextureName_SpriteNumber”. This quickly becomes hard to work with, and the Renamer tool seeks to remedy this problem.

User Interface



When opened the tool will show a User Interface like pictured on the right. All fields except for the Spritesheet field will remain disabled until a Sprite has been chosen for the Spritesheet field. Once chosen, all other fields become enabled, as well as the Rename button at the bottom.
The Selector dropdown allows you to change between Selectors to define Sprite sets within your sheet. Underneath the Selector dropdown the configuration options for the chosen Set Selector are drawn.
The Sprite Count label displays the amount of Sprites within the chosen Spritesheet.
Sprites Name is the pattern that the user wants to rename all the Sprites in the Spritesheet with. After configuration press the Rename button to rename all Sprites in the Spritesheet.

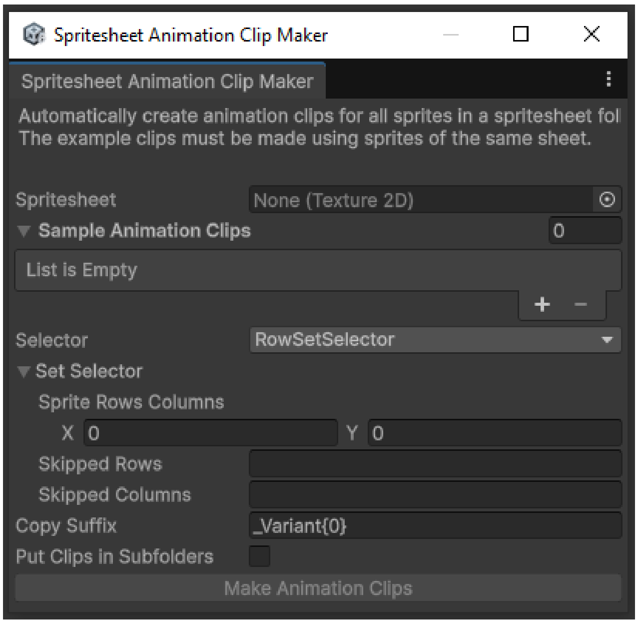
Sprites Name

The Sprites Name pattern will be used to name each Sprite within the Spritesheet. This pattern can use each Sprite’s setIndex and setSubIndex during the naming process. An example pattern “Character_{setIndex}_Pose_{setSubIndex}” will result in Sprites being named Character_0_Pose_0, Character_0_Pose_1, Character_0_Pose_2, Character_1_Pose_0, Character_1_Pose_1, Character_1_Pose_2, and so on.

Spritesheet Animation Clip Maker

Spritesheet Animation Clip Maker is a tool to quickly make new AnimationClips using all Sprites within a given texture2D Spritesheet, using previously made AnimationClips as a base. It is meant to rapidly make variations of existing Animations for different Sprite Sets.

User Interface



When opened the tool will show a User Interface like pictured on the right. The “Make Animation Clips” button will be disabled unless a Spritesheet and one or more Sample Animation Clips are chosen. The Spritesets within the chosen Spritesheet will be used for making new Animation Clips, whereas the Sample Animation Clips are the animations that will be duplicated and their Sprites replaced with the new Spritesets.
The Selector dropdown allows you to change between Selectors to define Sprite sets within your sheet and the Sample Animation Clips. Underneath the Selector dropdown the configuration options for the chosen Set Selector are drawn.
The Copy Suffix is a suffix to be applied on each newly created Animation Clip.
The Put Clips in Subfolders tickbox places the newly created Animation Clips in subfolders next to the Sample Animation Clips’ locations.

SetSelectors

SetSelectors are used both to verify each Sample Animation Clip only uses Sprites from a single Sprite Set, and to define which Sprite Sets in the Spritesheet to use for making new Animation Clips. Sample Animation Clips that don’t use Sprites or use Sprites that don’t fit within a single Sprite Set are skipped and logged with a warning.

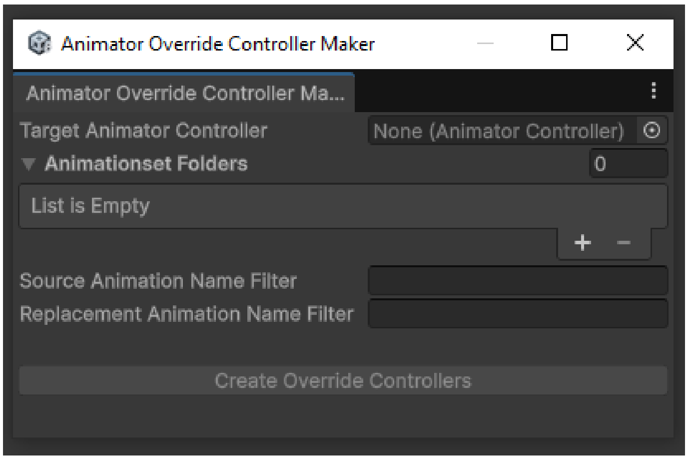
Copy Suffix

The Copy Suffix will be appended to each newly created Animation Clip’s name. This suffix can use the SetIndex of each Sprite Set for which an Animation Clip is made. The SetIndex’s position can be marked with “{0}”.

Animator Override Controller Maker

Animator Override Controller Maker is a tool to quickly make Animator Override Controllers for sets of AnimationClips that match the AnimationClips of the original AnimatorController. It can be used with Animations made through the Spritesheet Animation Clip Maker to quickly set up Animator Override Controllers for each Sprite Set within the Spritesheet.

User Interface



When opened the tool will show a User Interface like pictured on the right. The "Create Override Controllers" button will be disabled until a Target Animator Controller has been chosen, one or more Animationset Folders has been chosen, and a Source Animation Name Filter and a Replacement Animation Name Filter have been provided.
The Animationset Folder entries point to folders in the project containing a set of Animation Clips for which an Animator Override Controller must be made. Each entry can be filled in manually with a project-relative string, or a path can be selected using the button at the end of each entry.
The Source Animation Name Filter and Replacement Animation Name Filter are used to match the names of Animation Clips in the Target Animator Controller with the Animation Clips in the replacement sets respectively.
After configuration press the Create Override Controllers to create new Animation Override Controllers at the same location of the Target Animator Controller. New Controllers are made with a SetName matching the SetName of the replacement Animation Clips.

Source Animation Name Filter

The Source Animation Name Filter is used to define which part of an Animation Clip's name in the Target Animator Controller's Animation Clips has to be compared for replacement. All characters captured by a {compare} capture group will be compared. Literals can be written as-is, and greedy wildcards can be marked with "*". As an example, the Name Filter "Anim_*_{compare}_" will match with the animation "Anim_Character_Pose_1.anim" and capture "Pose" as the part to compare.

Replacement Animation Name Filter

The Replacement Animation Name Filter is used to define which part of an Animation Clip's name in the Animation Clips found in the Animationset Folders has to be compared with the

Target Animator Controller's Animation Clips. All characters captured by a {compare} capture group will be compared. Literals can be written as-is, and greedy wildcards can be marked with "*". A Set Name can be captured with {setName}, and will be used as a suffix for the created Animator Override Controller. As an example, the Name Filter "Anim_*_{compare}_{setName}." will match with the animation "Anim_Character_Pose_Variant1.anim" and capture "Pose" as the part to compare, and "Variant1" as the Set name.

SetSelectors

RowSetSelector

The RowSetSelector defines sets of Sprites within a Spritesheet as single rows within the sheet. Each Sprite will be given its row number as its setIndex, and its column number as its setSubIndex. For configuration it must be given the amount of rows and columns, including empty ones, within the Spritesheet. Empty rows and columns can be listed in the Skipped Rows and Skipped Columns fields, separated with commas. Ex: Skipped Rows 1,5,8 will skip rows 1, 5, and 8. Rows and columns are counted from 0.

ColumnSetSelector

The ColumnSetSelector defines sets of Sprites within a Spritesheet as single columns within the sheet. Each Sprite will be given its column number as its setIndex, and its row number as its setSubIndex. For configuration it must be given the amount of rows and columns, including empty ones, within the Spritesheet. Empty rows and columns can be listed in the Skipped Rows and Skipped Columns fields, separated with commas. Ex: Skipped Columns 1,5,8 will skip columns 1, 5, and 8. Rows and columns are counted from 0.

Custom SetSelectors

Custom SetSelectors can be implemented by creating subclasses from com.SolePilgrim.Unity.Editor.SpriteSheetTools.SetSelector.
Note: Custom SetSelectors are not automatically registered by any tools using them and must be added by updating the source code of each tool. This will be remedied at a later point.